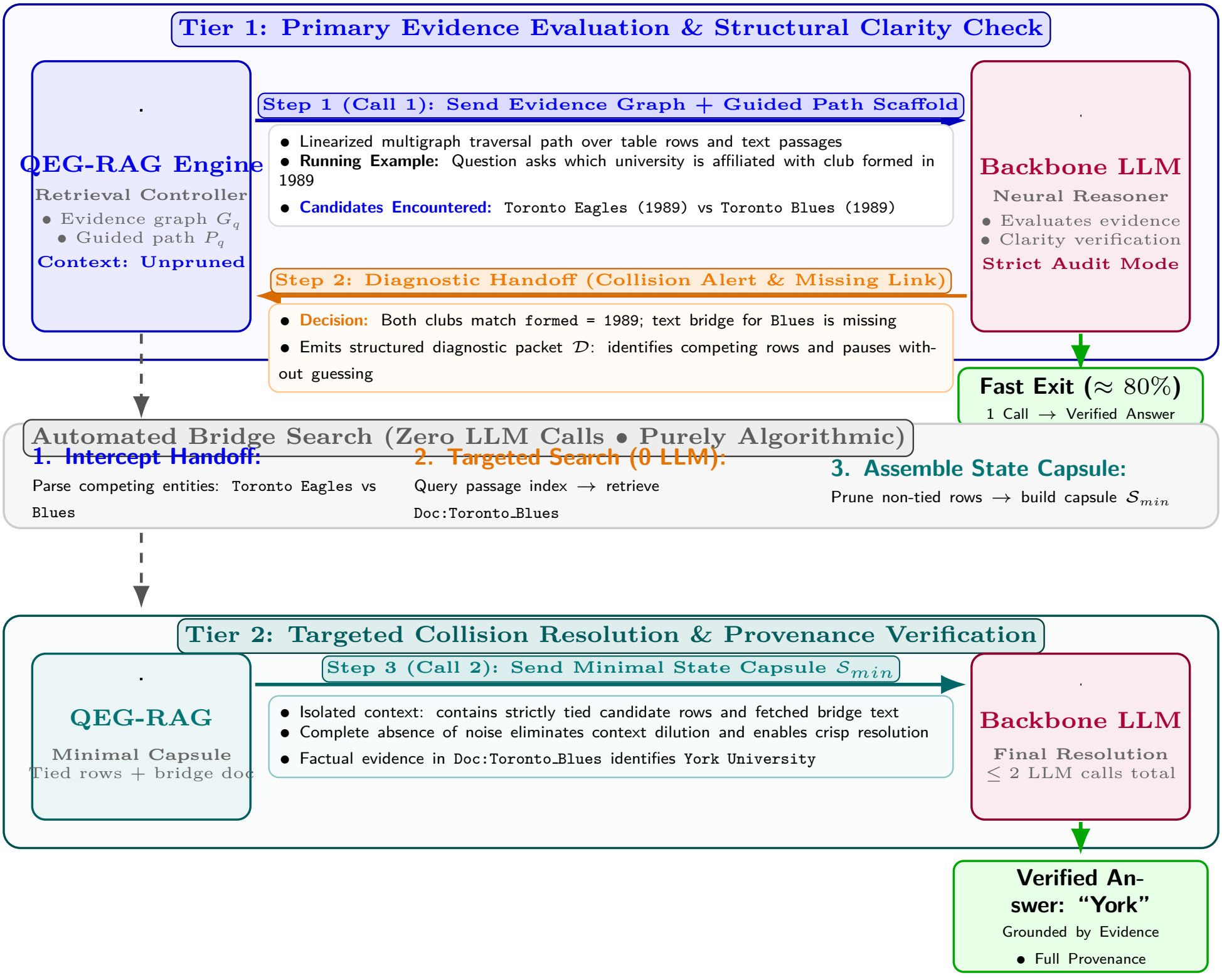

Tier 1: Primary Evidence Evaluation & Structural Clarity Check
QEG-RAG Engine
Retrieval Controller
• Evidence graph G_q
• Guided path P_q
Context: Unpruned
Step 1 (Call 1): Send Evidence Graph + Guided Path Scaffold
• Linearized multigraph traversal path over table rows and text passages
• Running Example: Question asks which university is affiliated with club formed in 1989
• Candidates Encountered: Toronto Eagles (1989) vs Toronto Blues (1989)
Backbone LLM
Neural Reasoner
• Evaluates evidence
• Clarity verification
Strict Audit Mode
Step 2: Diagnostic Handoff (Collision Alert & Missing Link)
• Decision: Both clubs match formed = 1989; text bridge for Blues is missing
• Emits structured diagnostic packet $\mathcal{D}$: identifies competing rows and pauses without guessing
Fast Exit (≈ 80%)
1 Call → Verified Answer
Automated Bridge Search (Zero LLM Calls • Purely Algorithmic)
1. Intercept Handoff:
Parse competing entities: Toronto Eagles vs Blues
2. Targeted Search (0 LLM):
Query passage index → retrieve Doc:Toronto_Blues
3. Assemble State Capsule:
Prune non-tied rows → build capsule $\mathcal{S}_{min}$
Tier 2: Targeted Collision Resolution & Provenance Verification
QEG-RAG
Minimal Capsule
Tied rows + bridge doc
Step 3 (Call 2): Send Minimal State Capsule $\mathcal{S}_{min}$
• Isolated context: contains strictly tied candidate rows and fetched bridge text
• Complete absence of noise eliminates context dilution and enables crisp resolution
• Factual evidence in Doc:Toronto_Blues identifies York University
Backbone LLM
Final Resolution
≤ 2 LLM calls total
Verified Answer: "York"
Grounded by Evidence
• Full Provenance